Reference Manual Cross-Validation Analysis Report

AQUABC_Reference_Manual.md & ESTAS_Reference_Manual.md
vs. Erturk et al. (2023) Ecological Modelling 486, 110509
vs. Actual Fortran Source Code

This document presents a systematic cross-validation of the two ESTAS-AQUABC reference manuals against the published paper by Erturk et al. (2023) and the actual Fortran source code. Each finding compares what the manual says, what the paper says, and what the code actually does.

20 findings were identified across four severity levels:
- CRITICAL (2): Errors that could mislead users or cause integration failures.
- MODERATE (5): Inaccuracies or significant omissions.
- LOW (7): Minor discrepancies or stale information.
- INFO (6): Expected differences or confirmations.

Documents analysed:
[1] AQUABC_Reference_Manual.md (410 lines) - Ecological model documentation
[2] ESTAS_Reference_Manual.md (525 lines) - Transport framework documentation
[3] Erturk et al. (2023) Ecological Modelling 486, 110509 - Primary model description paper
[4] SOURCE_CODE/ - All Fortran source files (ground truth)

Date: February 2026

Executive Summary

Total findings: 20 | Critical: 2 | Moderate: 6 | Low: 8 | Info: 4

ID	Severity	Title
F-01	CRITICAL	State-Variable Index Ordering Mismatch
F-02	CRITICAL	Allelopathy Variable Indices Wrong in AQUABC Manual
F-03	MODERATE	Fortran Standard: Paper Says 2003, Manuals Say 90/95
F-04	MODERATE	RK2 Solver Described as 'Future' -- Actually Implemented
F-05	MODERATE	Constant Category Ranges Incomplete in AQUABC Manual
F-06	LOW	Sediment Constant Count: Manual Says 170, Code Says 171
F-07	LOW	mod_GLOBAL.f90 Line Count: Manual Says 286, Actual Is 285
F-08	INFO	Paper State Variable Count (21) vs Code (32)
F-09	MODERATE	Synthesizing Unit vs Liebig Minimum: Nuanced Reality
F-10	INFO	Macroalgae Model: In Manual, Not in Paper
F-11	LOW	Transport Equation Terminology Differences
F-12	INFO	Paper Constant Count (183) vs Code (323+)
F-13	LOW	Zooplankton Switching Power Hardcoded but Undocumented
F-14	MODERATE	AQUABC Manual Missing CTMI Parameter Name Clarification
F-15	LOW	ESTAS Manual Missing Repeat-Cycle Spin-Up Details
F-16	MODERATE	Hypoxia Three-Regime System Underdocumented
F-17	LOW	Nostocales Naming Inconsistency in Code

F-18	INFO	ESTAS Manual Correctly Documents 7 Derivative Components
F-19	LOW	CO2SYS K1K2 Constants Selection Undocumented
F-20	LOW	Eco-Exergy Output Disabled by Default -- Manual Doesn't Note This

Key Numerical Comparisons

Aspect	Manual	Paper	Code	Verdict
State variables	32 (manual)	21 (paper)	32 (code)	**Manual correct**
Pelagic constants	318 (manual)	183 (paper)	323 (code)	**All different**
Sediment constants	170 (manual)	N/A (paper)	171 (code)	**Off by one**
Transported vars	32 (manual)	21 (paper)	36 (code)	**Manual wrong**
Fortran standard	90/95 (manual)	2003 (paper)	90/95 (code)	**Manual correct**
Solver options	Euler only (manual)	N/A (paper)	Euler + RK2 (code)	**Manual understates**

Detailed Findings

F-01 CRITICAL State-Variable Index Ordering Mismatch

Manual says: AQUABC Reference Manual Table (Section 2.1) lists: index 5=DIA-C, 6=CYN-C, 7=OPA-C, 8=FIX-CYN-C, 9=ZOO-C, 10=DET-C, ... 20=Mn(II), 21=Mn(IV), 22=Fe(II), 23=Fe(III), 24=S(-II), 25=S(VI), 26=CH4, 27-28=NOST, 29-32=Allelopathy.

Paper says: Paper Table 1 lists 21 variables with a completely different numbering: 1=NH4, 2=NO3, 3=PO4, 4=DSi, 5=DO, 6=Dia-C, 7=Cyn-NOFIX, 8=Cyn-FIX, 9=OPA-C, 10=Zoo-C, ...

Code shows: aquabc_II_pelagic_svindex.f90 defines: 1=NH4, 2=NO3, 3=PO4, 4=DOXY, 5=DIA-C, 6=ZOO-C, 7=ZOO-N, 8=ZOO-P, 9=DET_POC, 10=DET_PON, 11=DET_POP, 12=DOC, 13=DON, 14=DOP, 15=CYN-C, 16=OPA-C, 17=DSi, 18=PSi, 19=FIX_CYN-C, 20=DIC, 21=ALK, 22=FE_II, 23=FE_III, 24=MN_II, 25=MN_IV, 26=CA, 27=MG, 28=S(+6), 29=S(-2), 30=CH4-C, 31=NOST_VEG_HET, 32=NOST_AKI.

Analysis:

Neither the manual NOR the paper match the actual code ordering. The manual claims index 6 = CYN-C but the code has ZOO-C at index 6. The paper claims index 6 = Dia-C but the code has Dia-C at index 5. Zooplankton C/N/P occupy indices 6-8 in the code, not 9-11 as either document suggests. CYN-C is at index 15, OPA-C at 16, FIX_CYN-C at 19 -- none matching the manual's or paper's listing. Furthermore, the manual claims indices 29-32 are allelopathic metabolites; the code has S(-II), CH4, and Nostocales stages at 29-32, with allelopathy at 33-36.

Recommendation:

The AQUABC Reference Manual state variable table must be completely rewritten to match the actual index assignments in aquabc_II_pelagic_svindex.f90. The paper's Table 1 uses a logical (not code) numbering which should be noted.

F-02 CRITICAL Allelopathy Variable Indices Wrong in AQUABC Manual

Manual says: AQUABC manual lists indices 29-32 as 'SEC-METAB 1-4' (allelopathic secondary metabolites).

Paper says: Paper does not mention allelopathy.

Code shows: mod_GLOBAL.f90: nstate=32, NUM_ALLOLOPATHY_STATE_VARS=4. Allelopathy variables are indices 33-36, appended AFTER the 32 standard state variables. Indices 29-32 in the code are S(-II), CH4-C, NOST_VEG_HET, NOST_AKI.

Analysis:

The manual incorrectly maps allelopathy metabolites to indices 29-32, which actually hold S(-II), CH4, and Nostocales life-cycle stages. The total transported variables are 36 (32 standard + 4 allelopathy), not 32.

Recommendation:

Correct the manual table. Clearly state that the 32 standard state variables are followed by 4 allelopathic metabolites (indices 33-36) for a total of 36 transported variables.

F-03 MODERATE Fortran Standard: Paper Says 2003, Manuals Say 90/95

Manual says: ESTAS manual: 'Fortran 90/95'.

Paper says: Paper Section 2.1: 'Fortran 2003 was used'.

Code shows: No -std= flag in Makefile. Zero F2003/F2008 features found (no class(), abstract, type-extends, move_alloc, select type, or iso_fortran_env). All constructs are Fortran 90/95 compatible: module/use, type definitions, allocatable, where, SELECTED_REAL_KIND.

Analysis:

The paper's claim of Fortran 2003 is incorrect. The code uses exclusively Fortran 90/95 features. The manuals are accurate.

Recommendation:

Note this as a minor erratum in the paper. Both manuals already have the correct designation (Fortran 90/95).

F-04 MODERATE RK2 Solver Described as 'Future' -- Actually Implemented

Manual says: ESTAS manual Section 6: 'The infrastructure supports future addition of higher-order methods (e.g. Runge-Kutta 2)'.

Paper says: Paper does not discuss numerical solver.

Code shows: mod_SOLVER.f90: PELAGIC_SOLVER_NO == 2 triggers a FULLY IMPLEMENTED Heun's method (RK2) at lines 280-410+, with two-stage derivative calculation, averaged update, negative mass handling, and concentration clamping. Default is still Forward Euler (PELAGIC_SOLVER_NO = 1).

Analysis:

RK2 is not 'future' -- it is fully implemented and selectable. The manual understates the current capability.

Recommendation:

Update the ESTAS manual to document RK2 as an available solver option, with guidance on when to use it (e.g. stiff problems, larger time steps).

F-05 MODERATE Constant Category Ranges Incomplete in AQUABC Manual

Manual says: AQUABC manual lists 15 categories, last being 'Nostocales (276-298)'. Constants 299-323 are not accounted for in the category listing.

Paper says: Paper says 183 calibrated constants (Supplementary B).

Code shows: MODEL_CONSTANTS(1:318) array. Constants 299-306 are additional Nostocales parameters (akinete T/N thresholds). 307-309: POM dissolution saturation. 310-315: DON/DOP availability fractions. 316-318: phytoplankton-mediated mineralisation caps. 319-323: BETA photoinhibition (loaded via para_get_value, not the array).

Analysis:

The manual misses constants 299-323 (25 constants). The manual's category ranges also have overlaps and gaps (e.g. 'General 1-5' but category 2 starts at 5; 'POM dissolution 134-145' vs constant reference 134-146, etc.). The paper's '183' refers to calibrated subset only.

Recommendation:

Rewrite the constant category section to cover all 323 constants with accurate, non-overlapping ranges. Add the missing categories: POM dissolution saturation (307-309), DOM availability (310-315), phytoplankton mineralisation caps (316-318), and photoinhibition BETA (319-323).

F-06 LOW Sediment Constant Count: Manual Says 170, Code Says 171

Manual says: AQUABC manual: '170 parameters'.

Paper says: Paper does not specify sediment constant count.

Code shows: mod_GLOBAL.f90 line 40: NUM_SED_CONSTS = 171.

Analysis:

Off by one error in the manual.

Recommendation:

Correct the manual to state 171 sediment constants.

F-07 LOW mod_GLOBAL.f90 Line Count: Manual Says 286, Actual Is 285

Manual says: ESTAS manual source file table: 'mod_GLOBAL.f90 | 286'.

Paper says: N/A.

Code shows: wc -l returns 285 lines.

Analysis:

Off by one. Trivial but reflects stale documentation.

Recommendation:

Update line count in the ESTAS manual or remove exact line counts (they become stale quickly).

F-08 INFO Paper State Variable Count (21) vs Code (32)

Manual says: AQUABC manual correctly states 32 state variables.

Paper says: Paper Table 1 lists 21 state variables. Does not mention Fe(II/III), Mn(II/IV), Ca, Mg, SO4, H2S, CH4, or Nostocales life-cycle stages.

Code shows: nstate = 32 in mod_GLOBAL.f90.

Analysis:

The paper describes a simplified configuration with only 21 active state variables, used for the Curonian Lagoon case study. The additional 11 variables (Fe, Mn, Ca, Mg, S, CH4, Nostocales stages) exist in the code but were deactivated via flags (ADVANCED_REDOX_OPTION, DO_NOSTOCALES, etc.).

Recommendation:

This is expected and documented in the paper as a design choice. The AQUABC manual correctly lists all 32 (once the index ordering is fixed per F-01).

F-09 MODERATE Synthesizing Unit vs Liebig Minimum: Nuanced Reality

Manual says: AQUABC manual: SU 'replaces Liebig's minimum' for all phytoplankton groups.

Paper says: Paper Supplementary A implies Monod-type nutrient limitation.

Code shows: SU is used for N-P colimitation in all groups' non-fixing fraction. However: (1) Fixing fraction of N-fixing cyanobacteria retains Liebig min (because N-term is an inhibition switch, not colimitation); (2) SU of N-P is then cascaded with Si via SU for diatoms; (3) final growth limitation uses Liebig min(nutrient_lim, oxygen_lim); (4) Nostocales fixing fraction uses P-only limitation.

Analysis:

The manual's claim that SU 'replaces' Liebig is an oversimplification. SU replaces Liebig for N-P nutrient colimitation specifically, but Liebig min() is still used elsewhere (fixing fraction, oxygen-nutrient combination). There is no user-configurable switch between SU and Liebig.

Recommendation:

Revise the manual to explain that SU is used for N-P colimitation in the non-fixing pathway, while Liebig min is retained for the fixing fraction and for combining nutrient limitation with other factors (oxygen, light).

F-10 INFO Macroalgae Model: In Manual, Not in Paper

Manual says: AQUABC manual describes Macroalgae model with 6 state variables (Droop quota). ESTAS manual references mod_MACROALGAE.f90.

Paper says: Paper does not mention macroalgae at all. Table 2 notes some other models include 'Macroalgae or submerged aquatic vegetation' but AQUABC's entry shows 'No'.

Code shows: mod_MACROALGAE.f90 exists (372 lines) with 6 state variables and functional Droop quota kinetics.

Analysis:

The macroalgae module was developed after/alongside the paper and was not active in the Curonian Lagoon application. Curiously, the paper's Table 2 marks AQUABC as 'No' for macroalgae, which was accurate at the time of writing.

Recommendation:

This is an extension. The manuals correctly document it. No action needed, but the manuals could note that macroalgae was not used in the published case study.

F-11 LOW Transport Equation Terminology Differences

Manual says: ESTAS manual: 7 components -- advection, dispersion, settling, mass loads, mass withdrawals, kinetics, prescribed sediment fluxes.

Paper says: Paper Eq. 1: inflow from neighbours, outflow to neighbours, diffusion, settling from overlaying box, settling out of box, boundary forcing, kinetics.

Code shows: mod_SOLVER.f90: tot_deriv sums 7 arrays -- ECOL_ADVECTION_DERIVS, ECOL_DISPERSION_DERIVS, ECOL_SETTLING_DERIVS, ECOL_MASS_LOAD_DERIVS, ECOL_MASS_WITHDRAWAL_DERIVS, ECOL_KINETIC_DERIVS, ECOL_PRESCRIBED_SEDIMENT_FLUX_DERIVS.

Analysis:

The ESTAS manual matches the code exactly. The paper's formulation combines mass loads and withdrawals into 'boundary forcing' and splits settling into receiving/losing terms. Conceptually equivalent, but the paper omits prescribed sediment fluxes as a separate term.

Recommendation:

Add a note to the ESTAS manual mapping between the paper's Eq. 1 terminology and the code's derivative array names.

F-12 INFO Paper Constant Count (183) vs Code (323+)

Manual says: AQUABC manual: '318 parameters' (actually 323 with BETAs).

Paper says: Paper: '183 constants' in Supplementary Material B.

Code shows: nconst=318 array + 5 BETA named params = 323 in WCONST_04.txt. Plus EXTRA_WCONST.txt (allelopathy, ~30 more) and sediment constants (171).

Analysis:

The paper's 183 refers to the subset of constants that were actively calibrated for the Curonian Lagoon case study. The manuals correctly document the full model constant set. The 140 additional constants cover extended modules (redox, Nostocales, multi-EA anaerobic pathways, photoinhibition).

Recommendation:

Both manuals should clarify the distinction between 'calibrated constants' (paper's 183) and 'total model constants' (323+ pelagic, 171 sediment, allelopathy extras).

F-13 LOW Zooplankton Switching Power Hardcoded but Undocumented

Manual says: AQUABC manual describes Active Switching Model (Gentleman et al. 2003) but does not mention the switching power exponent value.

Paper says: Paper does not detail zooplankton feeding formulation.

Code shows: aquabc_II_pelagic_lib_ZOOPLANKTON.f90 line 194: switching power = 1.5 (hardcoded, not configurable via WCONST).

Analysis:

The switching power is an important model characteristic that affects food-web dynamics. It is hardcoded at 1.5 and cannot be changed without modifying source code.

Recommendation:

Document the switching power value (1.5) in the AQUABC manual zooplankton section and note that it is hardcoded.

F-14 MODERATE AQUABC Manual Missing CTMI Parameter Name Clarification

Manual says: AQUABC manual presents CTMI formula with T_min, T_opt, T_max but does not map these to the actual WCONST parameter names (OPT_TEMP_LR, OPT_TEMP_UR, KAPPA_OVER_OPT_TEMP).

Paper says: Paper does not explain parameter naming.

Code shows: aquabc_II_pelagic_auxillary.f90 comments document: Lower_TEMP = T_min, Upper_TEMP = T_opt, KAPPA_OVER_OPT_TEMP = T_max, KAPPA_UNDER_OPT_TEMP = unused.

Analysis:

The counter-intuitive naming (OPT_TEMP_LR sounds like 'lower optimal range' but is T_min; OPT_TEMP_UR sounds like 'upper optimal range' but is T_opt) is a persistent source of confusion. The manual should explicitly map parameter file names to their CTMI roles.

Recommendation:

Add a mapping table to the AQUABC manual CTMI section: OPT_TEMP_LR -> T_min, OPT_TEMP_UR -> T_opt, KAPPA_OVER_OPT_TEMP -> T_max, KAPPA_UNDER_OPT_TEMP -> unused.

F-15 LOW ESTAS Manual Missing Repeat-Cycle Spin-Up Details

Manual says: ESTAS manual mentions 'repeat cycles' for spin-up but gives no detail on how state is carried between cycles.

Paper says: Paper does not mention spin-up procedure.

Code shows: mod_SIMULATE.f90: the outer DO loop over NUM_REPEATS resets time to the simulation start but carries forward the final state variables as initial conditions for the next cycle. All output files continue appending.

Analysis:

The spin-up mechanism is straightforward (state carried forward, time reset), but this is important operational information for users.

Recommendation:

Document the spin-up procedure in the ESTAS manual: state variables are preserved across repeat cycles, time is reset to start, output continues appending.

F-16 MODERATE Hypoxia Three-Regime System Underdocumented

Manual says: AQUABC manual describes 'mortality enhanced under hypoxia' but does not detail the three-regime system.

Paper says: Paper mentions 'hypoxia stress on organisms' generically.

Code shows: All phytoplankton libraries implement three regimes: (1) DO > threshold: FAC_HYPOX = 1.0; (2) 0.1 < DO/threshold <= 1.0: FAC_HYPOX = THETA^(EXPON*(DO_thr - DO)); (3) DO/threshold <= 0.1: crash regime with capped mortality, growth and respiration zeroed.

Analysis:

The crash regime (regime 3) is a critical numerical safeguard that prevents mass-balance overshoot under near-anoxia. Neither the paper nor the manuals document it.

Recommendation:

Add full three-regime hypoxia documentation to the AQUABC manual with equations for each regime.

F-17 LOW Nostocales Naming Inconsistency in Code

Manual says: AQUABC manual uses NOST-VEG, NOST-HET notation.

Paper says: Paper does not mention Nostocales life cycle.

Code shows: Two different naming conventions coexist: FIX_CYN_HET_C_INDEX / FIX_CYN_AK_C_INDEX (old convention) and NOST_VEG_HET_C_INDEX / NOST_AKI_C_INDEX (new convention). Both refer to the same state variables (indices 31-32).

Analysis:

The dual naming is confusing but functionally harmless. The manuals use the newer NOST* convention, which is clearer.

Recommendation:

Deprecate the FIX_CYN_HET_C / FIX_CYN_AK_C names in favor of NOST_VEG_HET_C / NOST_AKI_C. Add deprecation comment.

F-18 INFO ESTAS Manual Correctly Documents 7 Derivative Components

Manual says: 7 components: advection, dispersion, settling, loads, withdrawals, kinetics, sediment fluxes.

Paper says: Paper Eq. 1: conceptually similar but different grouping.

Code shows: mod_SOLVER.f90 sums exactly these 7 derivative arrays.

Analysis:

The ESTAS manual is more accurate than the paper for this. The code implementation matches the manual exactly.

Recommendation:

No action needed. Manual is correct.

F-19 LOW CO2SYS K1K2 Constants Selection Undocumented

Manual says: AQUABC manual references CO2SYS CDIAC implementation but does not specify which K1K2 constant set is used.

Paper says: Paper cites Lewis & Wallace (1998) and van Heuven et al. (2011).

Code shows: aquabc_II_pelagic_model.f90: K1K2CONSTANTS = 4 (Mehrbach refit by Dickson & Millero 1987).

Analysis:

The choice of K1K2 constants affects pH calculation accuracy. Mehrbach/Dickson-Millero (option 4) is the standard choice for estuarine/coastal applications.

Recommendation:

Document the K1K2 constant selection (option 4 = Mehrbach/Dickson-Millero) in the AQUABC manual CO2SYS section.

F-20 LOW Eco-Exergy Output Disabled by Default -- Manual Doesn't Note This

Manual says: ESTAS manual describes eco-exergy diagnostics but does not mention that output is disabled by default.

Paper says: Paper does not mention exergy.

Code shows: mod_SIMULATE.f90: WRITE_PELAGIC_EXERGY_OUTPUT = 0 (disabled). mod_PELAGIC_EXERGY.f90 exists and is functional (123 lines).

Analysis:

Users would expect exergy output based on the manual but would not get it without changing the source code flag.

Recommendation:

Document how to enable exergy output in the ESTAS manual.

Appendix A: Correct State Variable Index Table

The following table shows the ACTUAL state variable indices from aquabc_II_pelagic_svindex.f90, which should replace the incorrect table in the AQUABC Reference Manual.

Idx	Code Name	Unit	Description
1	NH4_N	*mg N/L*	Ammonium nitrogen
2	NO3_N	*mg N/L*	Nitrate nitrogen
3	PO4_P	*mg P/L*	Orthophosphate phosphorus
4	DOXY	*mg O2/L*	Dissolved oxygen
5	DIA_C	*mg C/L*	Diatom carbon
6	ZOO_C	*mg C/L*	Zooplankton carbon
7	ZOO_N	*mg N/L*	Zooplankton nitrogen
8	ZOO_P	*mg P/L*	Zooplankton phosphorus
9	DET_PART_ORG_C	*mg C/L*	Detrital particulate organic C
10	DET_PART_ORG_N	*mg N/L*	Detrital particulate organic N
11	DET_PART_ORG_P	*mg P/L*	Detrital particulate organic P
12	DISS_ORG_C	*mg C/L*	Dissolved organic carbon
13	DISS_ORG_N	*mg N/L*	Dissolved organic nitrogen
14	DISS_ORG_P	*mg P/L*	Dissolved organic phosphorus
15	CYN_C	*mg C/L*	Non-fixing cyanobacteria carbon
16	OPA_C	*mg C/L*	Other planktonic algae carbon
17	DISS_Si	*mg Si/L*	Dissolved silica
18	PART_Si	*mg Si/L*	Particulate (biogenic) silica
19	FIX_CYN_C	*mg C/L*	N-fixing cyanobacteria carbon
20	INORG_C	*mg C/L*	Dissolved inorganic carbon
21	TOT_ALK	*meq/L*	Alkalinity
22	FE_II	*mg Fe/L*	Dissolved ferrous iron
23	FE_III	*mg Fe/L*	Particulate ferric iron
24	MN_II	*mg Mn/L*	Dissolved manganese(II)
25	MN_IV	*mg Mn/L*	Particulate manganese(IV)
26	CA	*mg Ca/L*	Calcium
27	MG	*mg Mg/L*	Magnesium
28	S_PLUS_6	*mg S/L*	Dissolved sulphate
29	S_MINUS_2	*mg S/L*	Dissolved sulphide
30	CH4_C	*mg C/L*	Dissolved methane
31	NOST_VEG_HET_C	*mg C/L*	Nostocales vegetative + heterocysts
32	NOST_AKI_C	*mg C/L*	Nostocales akinetes
33	SEC_METAB_1	*mg/L*	Allelopathic metabolite 1
34	SEC_METAB_2	*mg/L*	Allelopathic metabolite 2
35	SEC_METAB_3	*mg/L*	Allelopathic metabolite 3
36	SEC_METAB_4	*mg/L*	Allelopathic metabolite 4

White/alternating: core 21 variables (paper config). Blue tint: extended variables (code only). Orange tint: allelopathy variables (appended).

Appendix B: Recommended Corrections by Document

AQUABC Reference Manual

- F-01: Rewrite state variable index table (CRITICAL)
- F-02: Fix allelopathy variable indices 33-36, not 29-32 (CRITICAL)
- F-05: Complete constant category listing to cover 299-323 (MODERATE)
- F-06: Change '170' to '171' sediment constants (LOW)
- F-09: Clarify SU vs Liebig: SU for N-P non-fixing, Liebig elsewhere (MODERATE)
- F-13: Document zooplankton switching power = 1.5 hardcoded (LOW)
- F-14: Add CTMI parameter name mapping table (MODERATE)
- F-16: Document three-regime hypoxia system with equations (MODERATE)
- F-17: Note Nostocales naming: prefer NOST_* over FIX_CYN_* (LOW)
- F-19: Document CO2SYS K1K2 = 4 (Mehrbach/Dickson-Millero) (LOW)

ESTAS Reference Manual

- F-04: Document RK2 (Heun's method) as available solver, not 'future' (MODERATE)
- F-07: Update mod_GLOBAL.f90 line count or remove line counts (LOW)
- F-11: Add terminology mapping between paper Eq.1 and code derivative names (LOW)
- F-15: Document spin-up (repeat cycle) state carryover mechanism (LOW)
- F-20: Document how to enable eco-exergy output (LOW)

Paper Errata (Minor)

- F-03: Paper claims 'Fortran 2003'; code uses only Fortran 90/95 features
- F-08: Paper lists 21 state variables; code has 32 (paper describes subset -- acceptable)
- F-12: Paper says 183 constants; code has 323+ (paper describes calibrated subset -- acceptable)

Appendix C: References Consulted

[1] Erturk, A., Sakurova, I., Zilius, M., Zemlys, P., Umgiesser, G., Kaynaroglu, B., Pilkaityte, R. & Razinkovas-Baziukas, A. (2023) Development of a pelagic biogeochemical model with enhanced computational performance by optimizing ecological complexity and spatial resolution. Ecological Modelling, 486, 110509.

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